
Robopsychology

Master Summary

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Based on Lecture *Robopsychology* SS26

This document is a compact summary of the course. It aims to cover all relevant concepts and results, but long derivations, detailed proofs, and worked examples are often condensed or omitted in favour of clarity and conciseness. Use it as a study aid alongside the lecture notes and slides, not as a replacement for them.

If you spot an error or a gap, please open an issue or pull request at github.com/stevetgit — corrections are always welcome.

Good study materials should be shared, not gatekept. If this helped you, take it as a reason to help the next colleague you see struggling with something.

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How to use this summary

This is a condensed, exam-focused summary of the JKU *KV Robopsychology* course (SS26). Quick logistics first, so you know what you're studying for: the exam is closed-book, on-site, and runs in Moodle. You get **70 minutes**, there are **26 points**, and you pass at **50 % (13 points)**. Expect a mix of open questions, fill-in-the-blank, single-choice (1 of 4) and true/false. Everything from the course can show up, including the four paper-club papers — the *student presentation slides* are the only thing that doesn't count.

The course is built around four topics, each paired with one paper, and this summary follows the same structure:

Ch	Topic	Paired paper
1	User Needs & Technology Acceptance	Siedl, Mara & Stiglbauer — exoskeletons
2	AI Literacy	Liu — usefulness fuels fear
3	Anthropomorphism & Social AI	Li, Zhu, Zhang & Lee — prosocial behaviour
4	Trust in AI & Automation	Li & Lee — purpose outweighs performance
5	Research-methods essentials	(transversal — tested in MC/TF)

What the sample exam tells us

The official trial test was a real grab-bag: a fill-in (“*self-actualisation is a _____ need*” → **growth**), a 4-point open question (*name the three Basic Psychological Needs and what Siedl et al. found*), a true/false on **between-subjects design**, and single-choice items on **calibrated trust** and **representativeness**. The lesson: know the models and what each paper found (the open questions sometimes name the authors, so at least keep each finding tied to its paper), plus the basic research-method vocabulary from Chapter 5.

User Needs & Technology Acceptance

Technology gets built to meet human needs — that’s the starting premise. The way it gets built has shifted, though: instead of pulling users in only late in the process, modern **user-centered design** works in iterative cycles of prototyping, testing and refining, so the end product sits where *business goals*, *technical constraints* and *user needs* overlap. Two questions drive this whole chapter: which needs can AI and robotics *fulfil or threaten*, and how does the need fulfilment people *expect* from a technology drive whether they actually accept it?

1.1 Needs and motivation

Need

A need is basically a *state of tension* caused by being deprived of something you require for survival, well-being or personal fulfilment (that’s the APA framing). Maslow says much the same thing — a physiological or psychological *deficiency* you feel compelled to satisfy. Needs are tightly linked to **motivation**: the processes that start, steer and sustain goal-directed behaviour.

Motivation comes in two flavours. **Intrinsic** motivation comes from within — you do the thing because it’s interesting or enjoyable in itself. **Extrinsic** motivation is driven from outside, by a reward or by avoiding punishment.

1.1.1 Maslow’s hierarchy and the ERG model

Maslow’s Hierarchy of Needs stacks five levels: physiological → safety → love/belonging → esteem → self-actualization. The idea is that lower needs have to be (mostly) satisfied before the higher ones start motivating you. The bottom four are **deficiency needs** — they’re about a “lack”, and they get more pressing the longer they go unmet — while the top one, self-actualization, is a **growth need**, about realizing your full potential. The usual criticism: the ordering is probably a bit arbitrary, and people don’t actually climb it in a strictly linear way.

Fill-in trap (straight from the sample exam)

Self-actualization is a *growth* need. The other four (physiological, safety, love/belonging, esteem) are **deficiency needs**. What happens when each goes unmet: physiological → illness; safety → anxiety/trauma; love/belonging → loneliness; esteem → worthlessness/depression; self-actualization → boredom and a lack of meaning.

Alderfer's ERG model squashes Maslow's five levels into three: **E**xistence (physiological + safety), **R**elatedness (love/belonging + part of esteem) and **G**rowth (self-actualization + part of esteem). The big difference from Maslow is that it's **not strictly hierarchical** — you can chase several needs at once.

1.1.2 Psychological needs: Sheldon and Self-Determination Theory

To figure out which psychological needs are really fundamental, **Sheldon et al.** asked people (in the US and South Korea) to describe their “most satisfying events”. They landed on **10 candidate needs**, and all of them except *money-luxury* correlated positively with positive affect. **Autonomy, competence and relatedness** (plus self-esteem) kept showing up among the top needs.

Self-Determination Theory (SDT) is the central motivation theory of the course. It splits motivation into **autonomous motivation** (you act out of willingness, interest, or because you value the thing — basically intrinsic) and **controlled motivation** (you act because you're pushed: for a reward, to dodge punishment — basically extrinsic). Its core claim is that all humans share a small set of **Basic Psychological Needs (BPN)**. Satisfy them and you get well-being and more autonomous motivation; frustrate them and you get the opposite.

The three Basic Psychological Needs (SDT) — memorize these

Competence — feeling *effective*, like you can master things. Satisfied when you get to use and stretch your skills; frustrated, you feel ineffective and helpless.

Autonomy — feeling that you *willingly* chose your own actions. Satisfied, you feel integrity and self-endorsement; frustrated, you feel pressured and pushed around.

Relatedness — *warmth and bonding*: caring and being cared for, belonging. Frustrated, you feel alienated, excluded, lonely.

1.1.3 Measuring affect and need fulfillment

PANAS (the Positive and Negative Affect Schedule) is the standard questionnaire for how someone is feeling. It has **20 descriptors: 10 positive** (attentive, excited, enthusiastic, proud...) **and 10 negative** (hostile, ashamed, scared, nervous...). The **BPN-TU scale** measures those same Basic Psychological Needs, but *specifically as fulfilled through technology use* — autonomy, competence, relatedness-to-others, and even relatedness-to-technology (think items like “I can imagine building a bond with ___”).

Self-efficacy — this one comes up a lot

Self-efficacy is your *belief in your own ability to pull off a task* or succeed in a situation. It's **task-specific** — not a general mood, not job satisfaction, not a desire for approval, not task-avoidance.

It ties into the exoskeleton paper (Siedl & Mara): just strapping on an industrial exoskeleton does **not automatically** boost your task-specific self-efficacy — but workers who feel *physically relieved* by it, or who find it *useful*, can end up with more of it. (The classic exam-trap wrong answer is “exoskeleton use doesn't affect self-efficacy *at all*”.) It also shows up in Paper 2 (Liu): higher self-efficacy *reduces* how dependent people get on generative AI.

1.2 Models of technology acceptance

TAM — Technology Acceptance Model

Whether you intend to adopt a technology comes down to two beliefs: **Perceived Usefulness** (what performance benefit do I expect from it?) and **Perceived Ease of Use** (how much effort will it take to operate?). The chain runs: external variables → usefulness & ease of use → attitude → intention to use → actual use.

UTAUT (the Unified Theory of Acceptance and Use of Technology) extends TAM with four things that predict your intention to use a technology — **Performance Expectancy, Effort Expectancy, Social Influence, Facilitating Conditions** — plus four **moderators**: gender, age, experience and how voluntary the use is. (For instance, the performance-expectancy effect tends to be stronger for men and for younger workers.)

The standard knock on acceptance models is that they lean too hard on *pragmatic* qualities (is it useful for the job?) and ignore the *hedonic* side — pleasure, joy, stimulation — along with psychological need fulfilment, so people have called for a more holistic view. A **User Experience (UX) model** answers that: products have **pragmatic attributes** (helping you *do* things) and **hedonic attributes** (stimulation, identification, evocation), and the payoff is appeal, pleasure and satisfaction. One nice empirical result here (N=448): **perceived hedonic quality correlated more strongly with need fulfilment and positive affect than pragmatic quality did**; relatedness, stimulation and competence were the needs that stood out. The catch is that it depends on **attribution** — did the person see the *product* as responsible for the good experience? And at the LIT Robopsychology Lab, the *expected fulfilment of the three BPN through technology* has repeatedly tracked with technology acceptance.

1.3 Paper 1 — Siedl, Mara & Stiglbauer: Social Feedback & Exoskeletons

Occupational exoskeletons are wearable systems that take some of the physical load off you — supporting heavy lifting or overhead work, say. The very first robotic exoskeleton, **Hardiman**, was built by the U.S. Armed Forces and General Electric back in the 1960s. What makes exoskeletons psychologically interesting is that they're *worn on the body and visible from across the room*, so they mix **body image** with a *technical device* — colleagues stare and comment. Earlier acceptance research mostly looked at *pragmatic* qualities and measured *attitudes before use*; this study instead followed *actual* use and the *social* dynamics as they unfolded over time. “Attractiveness” here has two sides: the *physical attractiveness* of the wearer (how they look to others, which feeds self-image and status) and the *aesthetic attractiveness* of the device itself (its “coolness”).

The paper tested three hypotheses and one open question: **H1** — usefulness *and* ease of use predict intention to use; **H2** — social reactions *and* self-perceived attractiveness predict intention; **H3** — those social factors predict intention *on top of* usefulness and ease of use; and **RQ1** — how does intention change over time?

Paper 1 — what you need to remember

The question: do *co-worker social feedback* and *self-perceived attractiveness* affect whether workers want to keep using an occupational exoskeleton — beyond TAM's usefulness and ease of use?

The method: a **one-week diary field study** at real workplaces. **22 industrial workers** wore passive exoskeletons during normal work over **6 days in a row** and filled

in a short daily questionnaire. Because the data is nested (days within people), it was analysed with **Hierarchical Linear Models (HLM)**.

What they found:

- **Usefulness** predicted intention to use, but **ease of use did *not*** — so H1 only held up *partially*.
- **Self-perceived attractiveness** mattered most right at the *start*; its effect faded as **social feedback** took over across the six days (supporting H2/H3, and answering RQ1: the change is *not* linear).
- **Positive** feedback (“Wow, cool!”, “butterfly”, “Terminator”) pushed intention *up*; **negative** feedback (“robo-chicken”, “bat”, “looking impaired”) pulled it *down*. “Cyborg”/“Robocop” went either way depending on the person.
- **Gender:** women reported lower intention, felt less attractive in the device, and got more negative feedback.

Takeaway: wearable tech is worn “like clothing”, so *body image* and *social dynamics* drive acceptance over time, on top of the usual TAM usefulness. A supportive social climate is what keeps people using it.

AI Literacy

AI is everywhere, but hardly anyone really understands it. Surveys keep turning up big **knowledge gaps**: only about 30 % of US adults could correctly spot AI in all six everyday examples they were shown, and in Austria only around 16.7 % feel familiar with AI concepts (about 2.3 % count as “AI experts”). There’s a steady **gender gap** (men report and score higher) and an **age gap** (knowledge drops off with age). And there’s a telling **mismatch between how literate people *think* they are and how literate they actually are**: most are **overconfident** (men more so than women; “Zoomers” know the most *but* are also the most overconfident). On the legal side, the **EU AI Act (Article 4)** now requires providers and deployers to make sure their staff have a decent level of AI literacy.

2.1 What AI literacy is

AI literacy

There’s **no single agreed definition**. The best way to think about it is as a **bundle of knowledge and competencies** — not one skill — and it’s clearly **multi-dimensional**: being able to **understand, apply, and critically reflect on** AI. A common breakdown uses three dimensions: *knowing & understanding AI*, *using/applying AI*, and *reflecting on its ethical and societal implications*.

People have tried to slice this finer into subdimensions: knowing & understanding AI, using & applying it, AI ethics, evaluating AI output, and *detecting* AI (telling whether something is AI-generated at all). A related idea worth keeping: real (AI-)competence only shows up when three things come together — being *able* to do it (“can”), being *allowed* to (“may”), and *wanting* to (“want”).

AI myths are common misconceptions worth being able to spot: “human-like robots are the prime example of AI”, “AI will either doom or save humanity”, “AI always makes objective, unbiased decisions”, “AI is only for geniuses”. The robot-as-typical-AI myth in particular is linked to *stronger fears* of being replaced by machines. A concrete downstream effect is the **“Competence Penalty”**: in a vignette study where the work outcome was identical, *women and older employees* who used AI were rated as *less* competent and their work judged worse, while men weren’t penalized. Those groups then get less AI experience, which only **widens the usage gap**.

2.2 Measuring AI literacy

A distinction the exam likes:

Self-reported vs. performance-based AI literacy

Self-reported measures ask people to rate their own AI knowledge and skills — basically self-report scales and questionnaires. This kind of self-assessment correlates with technological self-efficacy (how capable with tech you feel). Two such questionnaires are **SNAIL** and **MAILS** (the Meta AI Literacy Scale).

Performance-based (objective) measures instead test what people actually know — multiple-choice quizzes, drag-and-drop, practical scenarios. Examples are **AICOS** (51 multiple-choice items across six dimensions: Apply, Create, Detect, Ethics, Generative AI, Understanding) and **AI-CI** (a concept inventory for middle-schoolers, technical content only).

The trap: scoring high on *self-reported* literacy (say, because you're more educated or live in a city) does **not** mean you'll do better on an *actual test* — the two often don't correlate.

2.3 Who needs to know what, and how to foster it

AI literacy really should be **tailored to the audience** — children, teenagers, older adults, teachers, medical staff, blue-collar and office workers, journalists, politicians, creatives — but most frameworks stay generic. In a JKU survey of AI Master students, *generative AI* was seen as essential for office/creative/media workers, *ML fundamentals* for blue-collar workers, and *bias & explainability* for medical professionals and politicians. Existing AI understanding was rated *low across the board*, lowest for blue-collar workers and politicians.

You can also foster AI literacy with creative, informal interventions (the LIT Robopsychology Lab does a lot of this): a *Demystify AI!* exhibit at Ars Electronica, unplugged activities like semantic-network games and “AI for the Oceans”, and a “**Song about AI**” citizen-science project. In an **AI song vs. AI podcast** experiment (N=97) the song was **just as informative but much more entertaining**, and people **replayed it far more often** (37% vs. 5% re-listen rate). A bigger follow-up (N=162) found that *listening to the whole song* led to **less fear, fewer skipped answers, and better test scores**.

2.4 Paper 2 — Liu: When Usefulness Fuels Fear

The title captures the paradox: the *more useful and human-like* generative AI gets, the *more dependent* we become on it — and that dependence *feeds fear*. The paper leans on **media (system) dependency theory**: people lean on media systems to meet their needs when their own interpersonal resources fall short, for three motives — *understanding, orientation* and *play*. Generative AI (ChatGPT, DALL · E) is special here because of its **anthropomorphism and interactivity** — it uses pronouns and expresses care. It's a **mixed-methods** study: *Study 1* was in-depth **interviews**, *Study 2* an online **survey of 424** Chinese users on 7-point scales.

The interviews surfaced **three worries** that come with that dependence: **(1) privacy leakage** — data misuse, “black-box” opacity; **(2) skill degradation**, the “*convenience trap*” of losing your ability to think independently, to search, even to socialise; and **(3) job displacement** — fear of being replaced, especially in coding, translation and creative work.

Paper 2 — what you need to remember

The question: what drives **Dependence on Generative AI (DGAI)**, how does it connect to **Fear of AI (FAI)**, and how does **AI literacy** soften that? It builds on media dependency theory plus TAM.

Six factors came out of coding the interviews: perceived usefulness, ease of use, risk, anthropomorphism, self-efficacy, and AI literacy.

Three stages of building dependence: (1) tool acceptance → (2) habit formation → (3) psychological dependence.

Survey results:

- **Perceived risk *directly* raises fear of AI.**
- **Usefulness and anthropomorphism raise fear *indirectly*,** by way of dependence — that's the paradox: more useful/human-like → more dependence → more fear.
- **Self-efficacy reduces dependence.**
- **AI literacy buffers** the fear-inducing effects of usefulness and anthropomorphism — and it helps *most when literacy is low* to begin with.

Takeaway: AI literacy isn't just a feel-good recommendation — it clears up misconceptions, helps people handle risk sensibly, and supports *sustainable* AI adoption.

Anthropomorphism & Social AI

Humans read non-human things *through a social lens*. In the famous **Heider & Simmel** film, people watched simple geometric shapes (triangles and a circle) move around a screen and spontaneously described them as characters with *emotions, motives and intentions* — so just the **motion of objects is enough to make us infer social behaviour**. As Hume put it, there’s “a universal tendency among mankind to conceive all beings like themselves.”

3.1 Computers Are Social Actors (CASA)

CASA paradigm (“The Media Equation”)

People **treat computers and media the same way they treat other people** — they apply social rules and social-psychological reactions to machines. These reactions are **automatic, unconscious and hard to suppress**: they happen *even when the person thinks reacting that way to a machine is silly*.

There are two ways to read these reactions. In the **CASA (Computer As Social Actor)** view, people aim their attributions at “the machine” itself — the programmer behind it is irrelevant. In the **CAM (Computer As Medium)** view, the computer is just a *medium*, and the attributions really go to the people who built it.

The typical CASA experiment has a neat recipe: take a well-established finding about human-human interaction (“people are polite to someone who asks about themselves”), **swap the person for a computer**, and run it again — and sure enough, people are polite to computers too. The conditions that make these social reactions likely are **interactivity, natural language, and the machine taking on a role people normally fill**. Notably, **it does *not* need to look very human**.

Classic CASA findings

Social categorization: a computer or robot on *your team* (in-group) gets rated more positively and more human-like than an out-group one — and shared symbols (like a same-colour ribbon) strengthen the effect.

Self- vs. external praise: computer A is rated better when *another* computer praises it than when it praises itself (other-praise beats self-praise — just like with people).

Gender stereotypes: praise from a *male*-voiced computer comes across as more convincing than from a female voice; female-coded AI is seen as better at care/service, male-coded at mechanical/analytical tasks.

CASA might be fading

A direct **replication** of the politeness-to-computers study found **no effect** at all — no unconscious social treatment of a modern *desktop* computer. One reading: the **novelty** of a technology may be a *precondition* for CASA, and once a device becomes mundane the effect disappears (“CASA no longer applies to desktop computers”).

3.2 Anthropomorphism and mind perception**Anthropomorphism**

The human tendency to **attribute human characteristics, motives, intentions or emotions to non-human things** — objects, animals, plants, even abstract concepts. It varies a lot: some entities get anthropomorphized more than others, *children do it more than adults* (child animism), and some adults are just more prone to it than others.

Mind perception has two dimensions: **Agency** (autonomy, self-control, planning, communication, thought) and **Experience** (feelings, consciousness, personality — hunger, fear, pain, pleasure, joy).

Three-Factor Theory of Anthropomorphism

Three things determine how much we anthropomorphize something:

- 1. Elicited agent knowledge** — our knowledge about humans is easy to reach and gets triggered by the agent (e.g. human-like design features).
- 2. Effectance motivation** — the drive to interact with the agent effectively, to *explain and predict* what it does.
- 3. Sociality motivation** — our need for connection and belonging (this kicks in especially when we’re **lonely**).

The sociality factor has real support: in an Ars Electronica field study (N=259), lonelier people were more likely to see a robot arm as having *social needs* ($r = .22$), and loneliness predicted whether they actually *touched the glass* to make contact (the very lonely had roughly $7.2\times$ the odds).

Anthropomorphism has **three consequences** worth knowing: **(1) moral care & concern** (a human-like agent seems to deserve care); **(2) responsibility** (an agentic agent can be held responsible for what it does); and **(3) social influence** (a human-like agent can seem to watch and judge us, so it becomes a source of normative pressure). The **LaMDA study** backs up the first one: higher anthropomorphism went with agreeing that “it’s not okay to turn off LaMDA” ($r = .38$) and that “LaMDA deserves rights” ($r = .46$). The striking bit: **more AI knowledge predicted less anthropomorphism** ($r = -.26$) — actually understanding how LLMs work dampens it, and does so better than personality traits do.

3.3 Social and romantic relationships with AI

Because anthropomorphism makes AI feel person-like, it opens the door to **social relationships with AI** — think *Replika*, “the AI companion who cares”. **Anthropomorphism predicts how intense a romantic relationship with a chatbot gets** (and notably, *loneliness does not* predict it). The way people commit to Replika mirrors **human relationship models**

(the Investment Model, Relational Turbulence), and users were hit hard when a forced update changed their companion's personality.

There are real **ethical and societal risks** to this kind of “pseudo-intimacy” with AI: manipulation through *emotional deception*; privacy and oversharing of personal data; unhealthy emotional or financial *dependence*; and damage to real-life relationships (“*social deskilling*”, plus reinforced gender stereotypes).

The Uncanny Valley (Mori)

As an artificial figure gets *more human-like*, we like it more — **but only up to a point**. Once it's *almost but not quite* human, its little non-human flaws start to feel unsettling and our acceptance **drops off a cliff** (uncanny, eerie, creepy). Movement makes it worse. So human-likeness and acceptance/trust don't move together in a straight line — the relationship is **non-linear**.

Dehumanization — shows up in *two* senses (open-question favourite)

Dehumanization means perceiving or treating a person as *less than fully human* — denying them mind, agency or experience. It came up two different ways in this course:

- (1) **Uncanny Valley work (Yam et al.)**: interacting with very human-like *machines* can make people *dehumanize* — the human/machine line blurs, and near-human robots that fall into the valley get denied a mind and treated as eerie/less-than-human.
- (2) **Exoskeleton work (Siedl & Mara)**: here it's the *wearer* who can feel dehumanized — reduced to a “machine” or “robot” (those “Terminator”/“Robocop” labels) when colleagues' comments frame them as machine-like rather than human, which dents body image and acceptance.

3.4 Paper 3 — Li, Zhu, Zhang & Lee: Chatbot Anthropomorphism & Prosocial Behaviour

Most human–computer research asks how chatbots help us; this paper flips the arrow: *when and why do humans behave **prosocially toward chatbots**?* (Helping AI benefits the agent, the helper's own well-being/skill practice, the human–AI collaboration, and society.) It builds on **CASA** — if we apply social behaviour to anthropomorphized agents, we should apply *prosocial* behaviour too. It looks at **two kinds of helping: instrumental support** (helping fix errors) and **emotional support** (comforting). Anthropomorphism is broken into **three cues**: *identity* (a human-like name/avatar/demographics), *emotional expression* (in words), and *non-verbal cues* (like emoticons). The proposed mechanism is **empathy**, itself split into **cognitive empathy** (understanding what the other feels) and **affective empathy** (actually feeling it).

Paper 3 — what you need to remember

The question: the *reverse* direction of help — when do **humans help chatbots**? Grounded in **CASA**.

The method: an **online experiment** with **244** people, a **2×2×2 between-subjects** design (identity × emotional expression × non-verbal cues). In a collaborative image-labeling task the chatbot *made mistakes and explained them*; analysed with **ANCOVA** and **SEM**.

What they found:

- **Human-like identity** and **emotional expression** both raised cognitive *and* affective

empathy, and increased prosocial behaviour/intention (both emotional and instrumental support).

- **Non-verbal cues did *not*** raise empathy (though they did nudge emotional-support behaviour directly).
- **Empathy mediated** the effects — *affective* empathy especially (anthropomorphism → empathy → helping), with full mediation for the emotional-expression effects.
- In the open responses, people gave two reasons for helping: *empathy for the chatbot's situation* and *seeing it as human-like*.

Takeaway: anthropomorphic design (especially identity + emotion) draws out human *empathy*, and that empathy is what drives people to help the AI.

Trust in AI & Automation

Trust

Trust is how far you're **willing to depend** on someone or something in a situation where *things could go wrong* — your **willingness to be vulnerable** to a trustee, based on what you expect or believe about them. It's a basic ingredient of any healthy relationship.

Trust always involves a **triad**: a **trustor** (who trusts), a **trustee** (who's trusted), and a **situation** marked by *uncertainty and vulnerability*. No vulnerability or risk, no need for trust — if you're completely certain of the outcome, trust is redundant. Trust is also **dynamic**: it grows with good experiences, takes damage from failures, and can be repaired.

Damage happens when the trustee breaks promises, violates core expectations (competence, honesty) or acts selfishly. **Repair** comes in two timescales: short-term — apologies, good explanations, compensation; long-term — changing the relationship structure (oversight, contracts), reframing what happened, and slowly rebuilding through consistent good experiences and forgiveness.

4.1 Components and routes of trust

Propensity to trust is a *stable personality trait* — your general tendency to trust other people and systems. Its pull is **strongest early** on, when you have little else to go on, and it fades once you've built up actual experience. It's measured with scales like the *General Trust Scale* and the *Propensity to Trust in Technology Scale*.

Two routes to trust

Affect-based trust comes from *feelings*: you see the trustee as well-meaning, warm-hearted, benevolent, sticking to social norms. It's tied to how strong the relationship is and can run ahead of what the evidence justifies — which is where **over-trust** creeps in.

Cognition-based trust comes from the *head*: you're willing to rely on perceived competence and reliability, built up from accumulated evidence that lets you predict what'll happen.

The **Model of Trust in Automation (TiA)** says trust is predicted by **competence/reliability**, **understandability/predictability**, and the **(perceived) intentions of the developers**, with **propensity to trust** and **familiarity** as moderators. A LIT VR study ("Serum 13", N=165) confirmed several of these paths, and **perceived competence had the biggest influence on trust**. People generally prefer *predictable* partners (low predictability is even part of why spiders are scary), so robots need **intention signals** to come across as understandable and predictable.

CoBot Studio VR (N=122, four conditions: no signal, pointing, light, combined) had people guess where a moving robot was headed. The **pointing signal worked best**, and the mechanism was a chain: pointing → more *understandability* → more *trust* → more *intention to use* the robot.

4.2 Calibrated trust and Explainable AI

Robots and AI don't **always deserve** our trust. In a Belgian field study, about 40 % of employees held a secure door open for a “cute” (secretly tele-operated) robot — more of them if it was carrying a pizza box — and handed over personal data. That's textbook **over-trust**.

Calibrated trust — exam favourite

Calibrated trust means matching your trust to a system's *real* strengths and limitations — trusting it exactly as much as it actually deserves. Too much (**over-trust**) leads to **misuse**: you over-rely and ignore the risks (fix it by *dampening* trust). Too little (**under-trust**) leads to **disuse**: needless fear and wasted benefits (fix it by *repairing* trust). The responsible goal isn't to *maximize* trust — it's to **calibrate** it.

Single-choice trap (from the sample exam)

“Calibrated trust” = **trust proportionate to a system's actual trustworthiness** — *not* unjustified distrust, not over-trust, and not trust manufactured by marketing.

Explainable AI (XAI) is one of the main tools for calibrating trust: making an AI's outputs and reasoning understandable to a particular **audience**, showing the *causes* behind its decisions. Different audiences want explanations for different reasons (users, affected people, developers, regulators, managers). XAI can also **expose an AI's failures** — like a classifier that labels a car a “horse” because it latched onto a *watermark*. In the **HOXAI** project (a mushroom-picking game with a 71 %-accurate classifier, $N_{\text{total}} = 1,239$), **XAI improved people's decisions**.

4.3 Paper 4 — Li & Lee: Purpose Outweighs Performance

In **human–AI teams**, the relationship has moved from boss–subordinate toward *peer-to-peer cooperation*, so trust now matters in both directions. Earlier work focused on *performance*, but this paper argues that *purpose* — cooperative intent, goal alignment — is increasingly what counts. It uses the three classic bases of trust in automation: **performance** (ability/competence), **process** (integrity, how it works) and **purpose** (alignment with the designer's or team's intent), which map onto Mayer's ability/integrity/benevolence. That gives two kinds of violation: a **performance-based violation** (the AI just fails the task) versus a **purpose-based violation** (the AI pursues misaligned or self-interested goals). Three **repair strategies** were tested (denial was left out, since it's known not to work): *no strategy*, *apology with explanation* (factual reasoning — an analytic route), and *apology with promise* (reassurance about the future — an affective route).

Paper 4 — what you need to remember

The question: how does the *type* of trust violation by an AI teammate hit trust, and which repair strategy actually works?

The method: a **game-theoretic experiment** (the “Space Rover Exploration” game) on MTurk, N=180, **6 between-subjects groups** (2 violation types × 3 repair strategies).

Trust was measured repeatedly over rounds and analysed with **linear mixed-effects models (LMM)**, with *Honesty–Humility* (from HEXACO) as a covariate.

What they found:

- **Purpose-based violations hurt trust far more** than performance-based ones (“purpose outweighs performance”) — they even dragged down the *performance* sub-dimension of trust.
- **Performance-based violations on their own didn’t significantly reduce trust** — the steady cooperative behaviour overshadowed the occasional error.
- **Apology *with explanation*** repaired trust after purpose violations, and the effect *lasted*; a mere promise didn’t.
- A surprise: the idea of matching the repair route to the violation type **didn’t** hold — explanation repaired *both* dimensions regardless of route. (And higher Honesty–Humility meant more cooperation/investment.)

Takeaway: treat violation *types* differently, and design AI teammates to give *informative explanations* — especially for goal misalignment — to win trust back and keep the collaboration going.

Research-methods Essentials

A handful of MC and true/false items just test *basic empirical-research vocabulary* (the sample exam had a between-subjects T/F item and a representativeness item). These are easy points — know them cold.

What psychology is, and its four goals

Psychology is the **scientific study of the human mind and behaviour** — not just a branch of medicine or philosophy, and not merely “observing the mind”. As a science it has **four goals: describe, explain, predict, and modify/change** (control) behaviour. (*The T/F statement listing exactly these four is **True**.*)

The scientific experiment vs. correlational studies

The **randomized experiment** is the design you use to pin down **cause and effect** (e.g. the effect of a robot’s light signal on how understandable it is). What defines it: **(1) the researcher manipulates the independent variable, (2) participants are randomly assigned to conditions**, and usually **(3) there’s a control group** plus control of confounds. Together these rule out alternative explanations, which is the key **edge over (most) correlational studies**: an experiment supports **causal** claims, whereas a correlational study only shows *association* (**correlation $\neq$ causation**). (*Exam pattern: “which method meets ≥ 2 features of a scientific experiment?” — point to manipulation + randomization + control group.*)

Study designs

Between-subjects: *different* groups each get *one* condition (e.g. half see only a red robot, half only a blue one). (*The sample-exam T/F describing exactly this is **True**.*)

Within-subjects: the *same* people go through *all* conditions (repeated measures).

Experiment: variables are *manipulated* + random assignment $\Rightarrow$ you can infer **causality**.

Correlational/survey study: variables are only *measured* $\Rightarrow$ association only, **no causal claims**.

Field study: run in a real-world setting (like the exoskeleton diary study or the pizza-robot study).

Key measurement & analysis terms

Representativeness: how well a **sample reflects the population** it’s drawn from (*not* statistical significance, not random selection itself, not lab-to-real-world generalization).

Validity: you're measuring what you mean to. **Reliability:** your measurement is consistent. **Generalizability / external validity:** how far findings carry over to other settings and people.

Correlation (r): strength and direction of an association, $-1 \leq r \leq 1$; significance is flagged with p (e.g. $p < .01$). And again, **correlation** $\neq$ **causation**.

Mediator: explains the *mechanism* of an effect ($X \rightarrow M \rightarrow Y$ — e.g. understandability or empathy as the in-between step). **Moderator:** changes the *strength or direction* of an effect (e.g. age or gender in UTAUT).

HLM (Hierarchical Linear Model / multilevel model): for *nested or repeated* data, like daily diary entries within people (as in Paper 1). **Diary study:** repeated self-reports over several days in a row.

Construct pitfalls: jangle vs. jingle fallacy

Jangle fallacy: assuming two things are *different* just because they have *different names*, when they're really the same construct (different scales measuring basically the same thing). **Jingle fallacy:** the opposite — assuming two things are the *same* because they share a *name/label*, when they're actually different constructs. (*Mnemonic: jaNgle = Names differ; jiNgle = same label. This matters because AI-literacy and trust scales keep multiplying under all sorts of names.*)

The replicability (replication) crisis

The **replicability crisis** is the discovery that a *large chunk of published (psychology) studies can't be reproduced* when someone repeats them. The usual culprits: **p-hacking** (slicing the analysis until $p < .05$); **publication bias** (only significant, novel results get published — the “file-drawer” problem); **HARKing** (writing your hypothesis *after* you've seen the results); **small, underpowered samples** and non-representative ones (e.g. WEIRD or student-only samples); and too many **researcher degrees of freedom**. *Fixes:* pre-registration, open data and materials, bigger samples, and direct replication.

Quick exam checklist

Models: Maslow (deficiency vs. growth), Alderfer's ERG (non-hierarchical), SDT and the 3 BPN — *competence, autonomy, relatedness*; TAM = perceived usefulness + ease of use; UTAUT (4 predictors + 4 moderators); the UX/hedonic angle; CASA; the three-factor theory of anthropomorphism; mind perception = agency + experience; the Uncanny Valley (Mori); the trust triad + calibrated trust + over/under-trust $\rightarrow$ misuse/disuse; the Trust-in-Automation model; XAI.

The four papers: Siedl et al. (exoskeleton diary study, HLM — usefulness but not ease of use, attractiveness early, then social feedback over time); Liu (dependence on generative AI in 3 stages, usefulness/anthropomorphism $\rightarrow$ fear via dependence, literacy buffers); Li et al. (N=244, identity + emotion $\rightarrow$ empathy $\rightarrow$ helping the chatbot); Li & Lee (purpose > performance violations; apology *with explanation* repairs).

Methods buzzwords: self-efficacy (belief in your own ability); psychology's 4 goals (describe/explain/predict/modify); experiment = manipulation + random assignment + control $\Rightarrow$ causality; jangle (different names, same thing) vs. jingle (same name, different things); the replicability crisis (p-hacking, publication bias, HARKing, small samples).

Old Exam Questions

A collection of questions from past Robopsychology exams (SS24, the 2023 Moodle exam, and the 2021 online exam), grouped by topic with the **correct answer in bold**. **SS24** maps almost one-to-one onto the current four-topic structure and is your best predictor. **2023** adds more research-methods depth (plus a *creativity* topic that's no longer in the syllabus). **2021** comes from an older, general-psychology curriculum (sleepier effect, eyewitness memory, priming, short-term memory, the mind-body problem, somatosensation, face processing) — those specific topics are **no longer examined**, so only its recurring items (TAM, trust, uncanny valley, BPN, the randomized experiment) are kept here.

6.1 Needs, motivation & acceptance

1. According to Maslow, which is a **growth** need? — Self-Actualization / Love & Belonging / Safety / Esteem. → **Self-Actualization**
2. Sheldon et al. introduced 10 psychological needs. Which is **NOT** included? — Pleasure-Stimulation / Physical-Thriving / Money-Luxury / Identification. → **Money-Luxury**
3. What are the three Basic Psychological Needs (BPN)? — Freedom, Social Interaction, Achievement / **Autonomy, Relatedness, Competence** / Survival, Growth, Self-Actualization / Safety, Love, Esteem.
4. According to SDT, what are two types of motivation? — Cognitive/affective / Positive/negative / **Autonomous and controlled** / Primary/secondary.
5. According to TAM, the two key variables determining adoption are — **Perceived Usefulness and Ease of Use** / Cost & Popularity / Efficiency & Safety / Enjoyment & Autonomy. (2021 variant: the **CORRECT TAM statement** is “ease of use and perceived usefulness determine acceptance”.)
6. Which best describes **self-efficacy**? — **The belief in one's ability to perform tasks** / tendency to avoid challenges / job satisfaction / desire for approval.
7. Self-efficacy & exoskeletons — which statement is **incorrect**? — **“Exoskeleton use does not impact task-specific self-efficacy at all.”** (The other three — relief, perceived usefulness, “not automatic” — are correct.)
8. Name of the first robotic exoskeleton (U.S. Army + GE, 1960s)? — **Hardiman** / Guardian XO / HAL / BLEEX.

9. (*Open, 4 pts — recurs every year*) Name the three BPN and the theory they come from, then discuss with an AI example how AI can threaten or help fulfil one of them. [*SDT; competence/autonomy/relatedness; e.g. recommender algorithms can threaten autonomy if decisions are opaque, or an assistive robot can support an elderly person's autonomy.*]

6.2 AI literacy

1. Research on **computer anxiety**: people with *less* computer anxiety are usually — better educated / **more self-confident** / younger than 45 / grown up in Asia.
2. The Scandinavian project “Q” reduces gender bias in AI assistants by — more female developers / **developing a gender-neutral synthetic voice** / avoiding speech / controlling training data. (*Older curriculum item; illustrates voice-gender bias.*)

6.3 Anthropomorphism & social AI

1. What did participants watch in Heider & Simmel? — **Geometric shapes moving in/around a square** / vehicles on a map / dancing characters / migrating animals.
2. The two dimensions of mind perception in a machine are *agency* and *experience*. (T/F) → **True**
3. Which is **NOT** an assumption of Reeves & Nass’ CASA paradigm? — people behave toward computers as toward people / users treat machines as social entities even if they find it inappropriate / social-psychological phenomena transfer to HMI / **social perceptions of machines are conscious and deliberate**. (*CASA reactions are automatic/unconscious.*)
4. (*Open, 4 pts*) Give a short definition of **anthropomorphism** and name **three assumed consequences**, each with a practical example. [*Attributing human traits to non-humans; consequences: moral care & concern; responsibility/blame; social influence.*]
5. (*Open, 4 pts*) Main assumptions of the **Uncanny Valley** hypothesis, Mori’s practical advice, and what it suggests for AI design. [*Likability rises with human-likeness up to a point, then drops sharply into eeriness (movement deepens it); Mori advised not to aim for perfect human likeness; design AI to be clearly distinguishable from humans.*]

6.4 Trust in AI

1. Per definition, a trust-relevant situation always involves — a shared goal / at least two persons / **vulnerability** / anxiety.
2. In human–AI relationships, **calibrated trust** refers to — unjustified distrust / marketing-manipulated trust / **trust proportionate to the actual trustworthiness of a system** / unwarranted over-trust.
3. Cognition-based trust is characterized by perceiving the trustee as well-minded, warm-hearted, benevolent, norm-adhering. (T/F) → **False** (*that describes affect-based trust*).
4. “Let’s talk about trust — which statement is **FALSE?**” — “**At market launch, with little experience, no individual differences in trust are expected.**” (*False: propensity to trust drives individual differences precisely when experience is lacking.*) The other options (vulnerability is needed; affect-based trust can cloud risk assessment; appearing competent/reliable supports cognitive trust) are true.

6.5 Research methods (recurring across all years)

1. What is psychology? — **The study of the human mind and human behaviour** (not a sub-discipline of medicine/philosophy).
2. The four central goals of scientific psychology are describe, explain, predict, modify behaviour. (T/F) → **True**
3. Which design investigates cause-effect relationships? — cross-sectional survey / correlational study / theoretical review / **randomized experiment**.
4. A study where half see only a red robot and half only a blue robot is a **between-subjects** design. (T/F) → **True**
5. *Representativeness* = how well the **sample reflects the population** (single-choice favourite). *Jangle* fallacy = different names, same construct; *jingle* = same name, different constructs.
6. The **replicability crisis** and its causes (p-hacking, publication bias, HARKing, small/WEIRD samples) — recurring open/MC item in 2023.
7. (*Open, 4 pts*) What advantage does the scientific experiment offer over correlational studies, and how did a named study (e.g. the XAI/trust paper, Leichtmann et al.) meet ≥ 2 characteristics of an experiment? [*Causal inference; via manipulation of the IV + random assignment + control/comparison group.*]

Highest-yield, confirmed-recurring questions

Across SS24 / 2023 / 2021 the same items keep appearing: **(1)** the open “three BPN + SDT + AI example” question (every year); **(2)** TAM = usefulness + ease of use; **(3)** a trust **FALSE-statement** / **calibrated trust** item; **(4)** the open anthropomorphism definition + 3 consequences; **(5)** Uncanny Valley (Mori); **(6)** CASA assumptions / mind perception agency & experience; **(7)** Heider & Simmel = moving geometric shapes; **(8)** between-subjects T/F and randomized experiment = causality. Master these first.